Hypertension: A Comprehensive Clinical Overview

1. Definition, Significance, and Global Epidemiology

Hypertension, defined clinically as persistently elevated arterial blood pressure at or above 130/80 mmHg (per the 2017 American College of Cardiology/American Heart Association guidelines) or 140/90 mmHg (per the European Society of Cardiology/European Society of Hypertension guidelines), is among the most common and consequential chronic medical conditions worldwide. It is characterized not by an acute, symptomatic presentation but by a prolonged, asymptomatic elevation in the force exerted by circulating blood against arterial walls. Because most patients experience no symptoms until catastrophic end-organ damage occurs, hypertension has earned the well-deserved designation of the "silent killer."

The global burden of hypertension is staggering. According to the World Health Organization, approximately 1.28 billion adults aged 30–79 years worldwide have hypertension as of the early 2020s, and that number is projected to rise substantially as populations age and urbanize. Roughly two-thirds of these individuals live in low- and middle-income countries. In the United States alone, the 2017 ACC/AHA reclassification elevated the estimated prevalence to approximately 46–47% of American adults — over 100 million people — compared with the prior estimate of about 32% under the older 140/90 threshold. Globally, hypertension is the leading attributable risk factor for cardiovascular disease mortality and disability-adjusted life years, surpassing tobacco use, high fasting plasma glucose, and obesity. It accounts for approximately 10.4 million deaths per year, predominantly through stroke and ischemic heart disease.

Despite the availability of effective and inexpensive treatments, control rates remain disappointingly low. Globally, only about 21% of women and 18% of men with hypertension have their blood pressure adequately controlled. In high-income countries, control rates are better but still suboptimal — in the United States, approximately 44% of adults with hypertension achieve adequate control. The "rule of halves" has been described in many health systems: roughly half of all hypertensives are unaware of their diagnosis, half of those who are diagnosed are not receiving treatment, and half of those on treatment are not controlled. These gaps translate into enormous preventable mortality and morbidity.

The disease disproportionately affects certain demographic groups. African American adults in the United States have among the highest rates of hypertension in the world — approximately 54–57% prevalence — with earlier onset, greater severity, and higher rates of complications such as stroke, end-stage renal disease, and heart failure. Hypertension prevalence increases steeply with age; nearly 70% of adults over 65 and over 80% of adults over 75 have hypertension. Men have higher prevalence than women before age 55, but this pattern reverses after the menopause transition, with postmenopausal women having higher rates than age-matched men.

2. Blood Pressure Physiology: Fundamentals

Blood pressure is defined as the force per unit area exerted by the blood on the walls of the arterial vasculature. It is expressed in millimeters of mercury (mmHg) and is characterized by two fundamental values: the systolic blood pressure (SBP) and the diastolic blood pressure (DBP). The systolic pressure is the peak arterial pressure reached during ventricular systole — the forceful contraction of the left ventricle ejecting blood into the aorta. The diastolic pressure is the minimum arterial pressure during ventricular diastole — the period of cardiac relaxation and ventricular filling between beats.

The fundamental hemodynamic relationship governing blood pressure is expressed by the equation: Blood Pressure = Cardiac Output × Systemic Vascular Resistance (BP = CO × SVR). This deceptively simple equation encapsulates two major physiological axes. Cardiac output (CO) is the product of stroke volume (the volume of blood ejected per beat, normally 60–100 mL) and heart rate (typically 60–100 beats per minute at rest), giving a normal resting CO of approximately 4–8 liters per minute. Systemic vascular resistance (SVR), also called total peripheral resistance, reflects the collective resistance offered by all systemic arterioles to blood flow, and is primarily governed by arteriolar tone, which is in turn regulated by neural, hormonal, and local factors. Either an increase in cardiac output, an increase in SVR, or both can produce hypertension.

Mean arterial pressure (MAP) is a clinically important derived value representing the average arterial pressure throughout the cardiac cycle. Because diastole occupies roughly two-thirds of the cardiac cycle at normal heart rates, MAP is not simply the arithmetic mean of SBP and DBP but is approximated by the formula: $MAP = DBP + 1/3(SBP - DBP)$, or equivalently $MAP = (SBP + 2 \times DBP) / 3$. A normal MAP is approximately 70–100 mmHg. MAP is the perfusion pressure that drives blood flow to most organ beds (except the coronary arteries, which are perfused predominantly during diastole). At a MAP below approximately 60 mmHg, autoregulatory mechanisms fail and critical organ ischemia begins to develop.

Pulse pressure is the difference between systolic and diastolic blood pressure: $PP = SBP - DBP$. A normal pulse pressure is approximately 40 mmHg (e.g., 120/80 yields a PP of 40). Pulse pressure is largely determined by stroke volume and arterial compliance (distensibility). With aging or atherosclerosis, the aorta and large arteries stiffen, reducing their ability to buffer each cardiac impulse, which causes the systolic pressure to rise and the diastolic to fall, widening the pulse pressure. An elevated pulse pressure ($>$60 mmHg) is an independent cardiovascular risk factor and is a hallmark of isolated systolic hypertension, which is the predominant pattern in older adults. Conversely, a narrowed pulse pressure can indicate reduced stroke volume, as in heart failure or cardiac tamponade.

3. Measurement of Blood Pressure

Accurate blood pressure measurement is the cornerstone of hypertension diagnosis and management, yet it is frequently performed incorrectly in clinical settings. The standard method for office measurement employs an inflatable cuff connected to a sphygmomanometer (mercury, aneroid, or electronic) placed around the upper arm at heart level. The cuff must be appropriately sized: the bladder should encircle at least 80% of the arm circumference, and width should be at least 40% of arm circumference. Use of an undersized cuff will overestimate blood pressure (a common error in obese patients), while an oversized cuff may underestimate it.

Proper technique for auscultatory measurement requires the patient to be seated quietly for at least 5 minutes in a chair with back support, feet flat on the floor, arm supported at heart level, without having smoked or consumed caffeine in the 30 minutes preceding the measurement. The cuff is inflated to at least 20–30 mmHg above the estimated systolic pressure and deflated slowly at 2–3 mmHg per second. The first Korotkoff sound (a tapping sound) corresponds to systolic pressure; its disappearance (Korotkoff phase V) defines diastolic pressure. At least two readings should be obtained at each visit, spaced 1–2 minutes apart, and the average recorded. On the first visit, blood pressure should be measured in both arms; a persistent inter-arm difference of $>$10–15 mmHg warrants investigation for subclavian artery disease or aortic coarctation.

Office blood pressure measurement is subject to significant variability and several specific phenomena that must be understood. White-coat hypertension (also called isolated office hypertension) refers to persistently elevated blood pressure measured in a clinical setting ($\geq$140/90 mmHg) with normal readings outside the office. It occurs in roughly 13–35% of patients classified as hypertensive by office measurement and carries a lower cardiovascular risk than sustained hypertension, though it is not entirely benign — patients with white-coat hypertension have a higher risk of developing sustained hypertension over time. White-coat effect (a less severe form) can be present in patients already on treatment, causing apparent treatment resistance in the office.

Masked hypertension is the converse phenomenon: normal or controlled blood pressure in the office (below 140/90 mmHg) but elevated readings outside the clinical setting. Masked hypertension, affecting approximately 10–20% of the general population and a larger proportion of treated patients, is clinically important because it carries a cardiovascular risk equivalent to sustained hypertension yet may be missed by office-only monitoring. Risk factors for masked hypertension include physical activity, anxiety, tobacco use, alcohol use, and poorly controlled daytime stress.

Ambulatory blood pressure monitoring (ABPM) is considered the gold standard for diagnosing hypertension and characterizing blood pressure patterns. It involves wearing a device for 24 hours (typically recording every 15–30 minutes during the day and every 30–60 minutes at night). ABPM provides daytime, nighttime, and 24-hour average values. Diagnostic thresholds for hypertension on ABPM are lower than office thresholds: 24-hour average $\geq$130/80 mmHg, daytime average $\geq$135/85 mmHg, and nighttime average $\geq$120/70 mmHg. Nocturnal blood pressure and the nocturnal "dip" pattern are particularly prognostically important. Normal individuals experience a 10–20% reduction in blood pressure during sleep ("dippers"). Reduced dipping ($<$10% reduction; "non-dippers") or reverse dipping (nighttime BP higher than daytime) are associated with increased target-organ damage and cardiovascular events. ABPM also enables calculation of blood pressure variability, which is an independent predictor of cardiovascular outcomes.

Home blood pressure monitoring (HBPM), using validated automated devices, provides an accessible and practical complement to office and ambulatory monitoring. Hypertension is diagnosed by HBPM when the average of multiple readings is ≥135/85 mmHg. Patients should be instructed to take readings in the morning (before taking medications) and in the evening for at least 4–7 days, discarding the first day, and averaging the remaining readings. HBPM has the advantage of improving patient engagement with their condition and reducing white-coat effect. Common measurement errors in routine practice include measuring over clothing, failure to rest before measurement, back or arm not supported, talking during measurement, using an improperly sized cuff, and reading only a single measurement.

4. Classification of Hypertension

The classification of blood pressure has undergone significant evolution, and two major international sets of guidelines currently coexist. The 2017 American College of Cardiology (ACC) and American Heart Association (AHA) guidelines substantially revised the blood pressure classification system compared with the previous JNC 7 framework. Under the 2017 ACC/AHA classification, blood pressure in adults is categorized as follows:

- **Normal:** Systolic less than 120 mmHg AND diastolic less than 80 mmHg. No pharmacologic intervention is indicated; lifestyle-promoting strategies are encouraged.
- **Elevated:** Systolic 120–129 mmHg AND diastolic less than 80 mmHg. This category (previously termed "prehypertension" under JNC 7 for readings 120–139/80–89) identifies individuals at heightened risk for developing hypertension who should pursue lifestyle modifications. No medications are recommended at this stage unless other high-risk conditions are present.
- **Stage 1 Hypertension:** Systolic 130–139 mmHg OR diastolic 80–89 mmHg. Lifestyle modification is indicated for all; pharmacologic treatment is recommended for those with clinical cardiovascular disease, a 10-year atherosclerotic cardiovascular disease (ASCVD) risk ≥10%, diabetes mellitus, or chronic kidney disease. For lower-risk Stage 1 patients, lifestyle changes alone are tried first.
- **Stage 2 Hypertension:** Systolic ≥140 mmHg OR diastolic ≥90 mmHg. Both lifestyle modification and pharmacologic treatment are recommended for all patients in this category.
- **Hypertensive Crisis:** Systolic greater than 180 mmHg and/or diastolic greater than 120 mmHg. Hypertensive crisis is subdivided into hypertensive urgency (no acute target-organ damage) and hypertensive emergency (with acute target-organ damage such as hypertensive encephalopathy, aortic dissection, acute pulmonary edema, or hypertensive nephrosclerosis).

In contrast, the 2018 European Society of Cardiology (ESC) and European Society of Hypertension

(ESH) guidelines retain the traditional 140/90 mmHg threshold for defining hypertension, classifying blood pressure as: Optimal (below 120/80), Normal (120–129/80–84), High-Normal (130–139/85–89), Grade 1 Hypertension (140–159/90–99), Grade 2 Hypertension (160–179/100–109), and Grade 3 Hypertension (≥180/110 mmHg). The ESC/ESH also categorize isolated systolic hypertension (systolic ≥140 with diastolic below 90), which is the predominant pattern in elderly patients. The differing thresholds between ACC/AHA and ESC/ESH guidelines mean that tens of millions of individuals are classified as hypertensive under American guidelines but only as "high-normal" under European guidelines, which has important implications for treatment decisions and public health messaging.

Isolated systolic hypertension (ISH) deserves special mention. ISH is defined as SBP ≥140 (or ≥130 under ACC/AHA) with DBP below 90 (or below 80). ISH is the most common pattern in adults over 60, affecting more than two-thirds of hypertensives in that age group, and results predominantly from age-related arterial stiffening. Despite older beliefs that elevated systolic pressure was a normal consequence of aging and did not require treatment, landmark clinical trials (SHEP, Syst-Eur, Syst-China) firmly established that treating ISH in the elderly reduces stroke and cardiovascular mortality.

5. Physiological Regulation of Blood Pressure

Blood pressure is regulated through a complex, multilayered system of neural, hormonal, and intrinsic vascular mechanisms that operate on timescales ranging from milliseconds (baroreceptor reflexes) to days and weeks (renal pressure-natriuresis). Understanding these mechanisms is essential for understanding how hypertension develops and how antihypertensive drugs work.

The Renin-Angiotensin-Aldosterone System (RAAS)

The renin-angiotensin-aldosterone system is arguably the most pharmacologically important blood pressure regulator. The cascade begins in the juxtaglomerular apparatus of the kidney, where specialized granular cells in the afferent arteriole secrete the enzyme renin in response to three major stimuli: reduced renal perfusion pressure (sensed by intrarenal baroreceptors), decreased sodium chloride delivery to the macula densa cells of the distal tubule, and sympathetic nervous system activation via beta-1 adrenergic receptors. Renin cleaves the plasma protein angiotensinogen (produced by the liver) into the decapeptide angiotensin I. Angiotensin I is itself largely biologically inactive but is converted to the octapeptide angiotensin II by angiotensin-converting enzyme (ACE), which is found predominantly in the pulmonary vascular endothelium but also in the kidney, heart, and brain.

Angiotensin II is a potent vasoconstrictor acting primarily on AT1 receptors, and it exerts multiple blood pressure-elevating actions: direct arteriolar vasoconstriction raising SVR; stimulation of aldosterone secretion from the adrenal zona glomerulosa; stimulation of antidiuretic hormone (ADH/vasopressin) release from the posterior pituitary; increased thirst and sodium appetite; enhancement of norepinephrine release from sympathetic nerve terminals; and direct stimulation of proximal tubular sodium reabsorption. Aldosterone, in turn, acts on mineralocorticoid receptors in the principal cells of the collecting duct to increase sodium reabsorption and potassium excretion through upregulation of the epithelial sodium channel (ENaC) and the sodium-potassium ATPase pump. The net effect of RAAS activation is increased sodium retention, expanded plasma volume, and elevated blood pressure.

The Baroreceptor Reflex

The baroreceptor reflex provides rapid, second-to-second blood pressure homeostasis. High-pressure baroreceptors (mechanoreceptors) are located in the carotid sinus at the bifurcation of the common carotid artery and in the aortic arch. These stretch-sensitive receptors fire at a rate proportional to arterial pressure and transmit signals via the glossopharyngeal nerve (CN IX) from carotid baroreceptors and the vagus nerve (CN X) from aortic baroreceptors to the nucleus tractus solitarius in the medulla. Increased firing inhibits sympathetic outflow and increases parasympathetic (vagal) tone, causing vasodilation, reduced heart rate, and reduced cardiac contractility — collectively lowering blood pressure. A drop in blood pressure has the opposite effect, rapidly activating the sympathetic nervous system to restore perfusion. In chronic hypertension, baroreceptors "reset" to the elevated pressure level, losing their effectiveness as a moment-to-moment buffer, which is one reason why chronic hypertension perpetuates itself.

The Sympathetic Nervous System

Sympathetic nervous system activation raises blood pressure through multiple mechanisms: increased heart rate and contractility (via beta-1 adrenergic receptors in the myocardium), arteriolar vasoconstriction (via alpha-1 receptors in vascular smooth muscle), increased renin release (via renal beta-1 receptors), and increased venous tone enhancing venous return and preload. Chronically elevated sympathetic nervous system activity, which has been demonstrated in early and established essential hypertension, contributes to both the initiation and maintenance of elevated blood pressure.

Natriuretic Peptides

The natriuretic peptide system counterbalances the RAAS and sympathetic nervous system. Atrial natriuretic peptide (ANP) is secreted by atrial cardiomyocytes in response to atrial wall stretch from volume expansion. Brain natriuretic peptide (BNP, actually produced by ventricular myocardium) is released in response to ventricular wall stress. Both peptides act on natriuretic peptide receptors to cause natriuresis (sodium and water excretion), vasodilation, suppression of renin and aldosterone secretion, and inhibition of sympathetic outflow. Their net effect is to lower blood pressure and reduce intravascular volume — a physiological counterpoint to RAAS activation.

Renal Pressure-Natriuresis

The kidney plays a central long-term role in blood pressure regulation through the pressure-natriuresis relationship. When arterial pressure rises, the kidney excretes more sodium and water (natriuresis and diuresis), reducing plasma volume and cardiac output back toward normal. Conversely, a fall in pressure reduces sodium excretion, expanding volume. This mechanism is considered the ultimate long-term blood pressure regulator because, in the absence of a resetting of the pressure-natriuresis curve, sustained hypertension is theoretically impossible. In essential hypertension, this curve is shifted rightward — the kidney requires a higher perfusion pressure to excrete a given sodium load — meaning that blood pressure must be elevated to achieve sodium balance. Factors that shift the curve include RAAS activation, sympathetic nervous system activity, and structural nephron loss.

Endothelial Factors

The vascular endothelium is a dynamic endocrine organ that continuously modulates vascular tone through the balanced release of vasodilatory and vasoconstrictive substances. Nitric oxide (NO), synthesized from L-arginine by endothelial nitric oxide synthase (eNOS), is the primary endothelium-derived relaxing factor. NO diffuses into vascular smooth muscle cells, activates guanylate cyclase, increases cyclic GMP, activates protein kinase G, dephosphorylates myosin light chains, and causes smooth muscle relaxation and vasodilation. Impaired NO bioavailability — due to reduced eNOS activity or increased oxidative degradation of NO by reactive oxygen species — is a key mechanism of endothelial dysfunction in hypertension. Endothelin-1 (ET-1) is a potent vasoconstrictor peptide also produced by endothelial cells, acting on ET-A receptors in smooth muscle to cause sustained vasoconstriction and promoting vascular remodeling. Prostacyclin (PGI2) from endothelium promotes vasodilation and inhibits platelet aggregation, while thromboxane A2 from platelets causes vasoconstriction. The balance between these factors is disturbed in hypertension, tilting toward a pro-vasoconstrictive, pro-inflammatory endothelial state.

6. Primary vs. Secondary Hypertension

Hypertension is broadly divided into primary (essential) hypertension and secondary hypertension based on whether an identifiable underlying cause exists. Primary hypertension accounts for approximately 90–95% of all hypertension cases and represents a complex, polygenic disorder with no single etiology. Secondary hypertension accounts for approximately 5–10% of cases but is disproportionately important to identify because it is often amenable to targeted therapy or cure, and because missing a secondary cause leads to suboptimal management.

Primary (Essential) Hypertension

Primary hypertension develops through the interaction of genetic susceptibility, lifestyle factors, and environmental influences over years. It typically manifests in the third to sixth decades of life and worsens progressively with aging. Genome-wide association studies (GWAS) have identified hundreds of genetic loci associated with blood pressure, each contributing small incremental effects. The heritability of blood pressure is estimated at 30–60%. Important pathophysiological contributors include increased sodium sensitivity of the kidneys (requiring higher pressure to achieve sodium balance), overactivation of the RAAS and sympathetic nervous system, endothelial dysfunction with reduced nitric oxide production, vascular stiffness with reduced large-artery compliance, low nephron number (Brenner hypothesis — individuals born with fewer nephrons have increased susceptibility to hypertension), and chronic low-grade inflammation.

Secondary Hypertension: Identifiable Causes

Secondary hypertension should be suspected when hypertension is severe, abrupt in onset, onset at a young age (particularly before 30 without risk factors), resistant to multiple medications, or accompanied by specific clinical clues. The major causes include:

- **Chronic Kidney Disease (CKD):** The most common secondary cause. Reduced functioning nephron mass impairs sodium excretion, activates the RAAS, and causes volume expansion. Hypertension is present in approximately 60–80% of CKD patients, and conversely CKD itself can result from hypertensive nephrosclerosis, creating a vicious cycle.
- **Renal Artery Stenosis (Renovascular Hypertension):** Narrowing of one or both renal arteries reduces perfusion to the affected kidney(s), activating the RAAS. Atherosclerotic renal artery stenosis (in older patients with coronary or peripheral arterial disease) and fibromuscular dysplasia (in young women, typically affecting the mid and distal renal arteries) are the two major subtypes. Clinical clues include flash pulmonary edema, recurrent hypokalemia, and a systolic-diastolic epigastric bruit.
- **Primary Aldosteronism (Conn's Syndrome):** Autonomous overproduction of aldosterone by one or both adrenal glands, independent of RAAS regulation. It is far more common than previously recognized, occurring in approximately 5–10% of hypertensive patients and up to 20% of those with resistant hypertension. It presents with hypertension, hypokalemia (though normokalemia is common), metabolic alkalosis, and suppressed plasma renin activity with elevated plasma aldosterone. Causes include bilateral adrenal hyperplasia (most common) and aldosterone-producing adenoma (Conn's adenoma). Screened with the aldosterone-to-renin ratio (ARR); if elevated, confirmatory testing with salt loading, fludrocortisone suppression, or captopril challenge is performed.
- **Pheochromocytoma:** Catecholamine-secreting tumor of the chromaffin cells, arising in the adrenal medulla (pheochromocytoma) or sympathetic ganglia (paraganglioma). It can cause episodic or sustained hypertension with classic paroxysms of headache, diaphoresis, and palpitations ("the three Ps"). It accounts for approximately 0.2–0.6% of hypertension cases but carries significant mortality if missed. Diagnosis involves 24-hour urine or plasma fractionated metanephrines and normetanephrines. Associated with genetic syndromes including MEN2 (RET mutation), von Hippel-Lindau disease, neurofibromatosis type 1, and succinate dehydrogenase mutations.
- **Cushing's Syndrome:** Chronic glucocorticoid excess from pituitary ACTH overproduction (Cushing's disease), adrenal adenoma, or exogenous corticosteroids. Glucocorticoids increase blood pressure through mineralocorticoid receptor activation, enhanced vascular responsiveness to vasoconstrictors, and increased cardiac output. Clinical stigmata include central obesity, "moon face," "buffalo hump," purple striae, muscle weakness, osteoporosis, and glucose intolerance.
- **Thyroid Disease:** Hypothyroidism classically causes diastolic hypertension through increased vascular resistance. Hyperthyroidism causes predominantly systolic hypertension

and tachycardia due to increased cardiac output.

- **Coarctation of the Aorta:** A congenital narrowing of the aorta, typically just distal to the left subclavian artery. It causes hypertension in the upper extremities with normal or low blood pressure in the lower extremities (the opposite of normal, where leg BP is typically 10–20 mmHg higher than arm). Classic findings include rib notching on chest X-ray (from dilated intercostal collateral vessels), weak femoral pulses, and an aortic systolic murmur. Usually diagnosed in childhood but can present in adults.
- **Obstructive Sleep Apnea (OSA):** Episodic upper airway obstruction during sleep causing repeated cycles of hypoxia, hypercapnia, arousal, and sympathetic activation. OSA is strongly associated with both hypertension and resistant hypertension, with each additional apneic episode per hour of sleep correlating with progressive increases in blood pressure. It is present in approximately 30–40% of hypertensive patients and up to 80% of those with resistant hypertension.
- **Drug-induced Hypertension:** Numerous medications can raise blood pressure, including nonsteroidal anti-inflammatory drugs (NSAIDs, which inhibit prostaglandin-mediated vasodilation and promote sodium retention), oral contraceptive pills (estrogen-mediated RAAS stimulation), sympathomimetics (decongestants, stimulants, cocaine, amphetamines), calcineurin inhibitors (cyclosporine, tacrolimus), erythropoiesis-stimulating agents, venlafaxine and other SNRIs, and excessive licorice (glycyrrhizinic acid inhibits 11-beta-hydroxysteroid dehydrogenase, allowing cortisol to activate mineralocorticoid receptors).

7. Risk Factors for Hypertension

Risk factors for developing hypertension are traditionally divided into non-modifiable and modifiable categories. Understanding these risk factors is essential for identifying high-risk individuals for targeted prevention and for counseling patients on lifestyle changes.

Non-modifiable risk factors include age (blood pressure rises progressively with age in most populations, particularly systolic pressure due to aortic stiffening), family history and genetics (having a first-degree relative with hypertension roughly doubles one's risk), race and ethnicity (Black individuals have earlier onset, higher prevalence, and more severe hypertension than White individuals, partly attributed to genetic variants affecting renal sodium handling), sex (men have higher prevalence before age 55; the gap narrows and reverses after menopause), and birth factors (low birth weight, premature birth, and intrauterine growth restriction are associated with reduced nephron number and higher adult blood pressure risk).

Modifiable risk factors carry particular importance because they offer opportunities for both prevention and treatment. Dietary sodium intake is one of the most extensively studied modifiable risk factors. The relationship between sodium and blood pressure is consistent across populations, though individuals vary in their degree of "salt sensitivity." Current average sodium intake in the United States is approximately 3,400 mg/day, substantially exceeding the recommended maximum of 2,300 mg/day. The INTERSALT study and multiple intervention trials have established that each 100 mmol/day reduction in urinary sodium is associated with a 1–2 mmHg reduction in systolic blood pressure in normotensive individuals and 5–6 mmHg in hypertensives. Obesity is a major risk factor; each 10 kg of weight gain raises systolic blood pressure by approximately 3 mmHg. The mechanisms include increased cardiac output, sympathetic nervous system activation, hyperinsulinemia with renal sodium retention, activation of the RAAS by adipose tissue, and physical compression of the kidneys by perirenal and retroperitoneal fat. Physical inactivity contributes to hypertension risk through its associations with obesity, insulin resistance, and reduced endothelial nitric oxide production. Aerobic exercise training consistently lowers blood pressure by approximately 5–8 mmHg systolic. Excessive alcohol consumption (more than 2 standard drinks per day in men, 1 in women) raises blood pressure through sympathetic activation and direct vascular effects; interestingly, low-to-moderate alcohol intake may have neutral or slightly protective effects on blood pressure. Tobacco/smoking, while not a direct cause of chronic hypertension, acutely raises blood pressure through nicotine-mediated catecholamine release and accelerates atherosclerosis and vascular damage, markedly amplifying the cardiovascular risk of hypertension. Chronic psychosocial stress, particularly from work stress, sleep deprivation, financial stress, and neighborhood violence, activates the sympathovagal axis and RAAS, with long-term adverse effects on blood pressure. Low dietary potassium intake is independently associated with higher blood pressure; potassium promotes natriuresis by reducing ENaC activity in the collecting duct.

8. Target-Organ Damage and Complications

The complications of hypertension arise from the direct mechanical effects of elevated pressure on vessel walls and the secondary structural changes (hypertrophy, fibrosis, endothelial dysfunction, atherosclerosis) that develop over years. The major target organs are the heart, brain, kidneys, eyes, and vasculature.

Cardiac Complications

The heart must continuously work against the elevated afterload imposed by hypertension. Over time, this pressure overload stimulates concentric left ventricular hypertrophy (LVH), defined as an LV mass index exceeding 115 g/m^2 in men and 95 g/m^2 in women (by echocardiography). LVH is detected in approximately 15–20% of hypertensive patients by echocardiography (and less sensitively by ECG, where a Sokolow-Lyon voltage criterion of S in V1 + R in V5/V6 > 35 mm is commonly used). LVH represents an independent cardiovascular risk factor above and beyond blood pressure itself. As LVH progresses, diastolic dysfunction develops — impaired ventricular relaxation and filling — leading to heart failure with preserved ejection fraction (HFpEF), one of the most common forms of heart failure in hypertensive patients. Ultimately, decompensation leads to systolic dysfunction and reduced ejection fraction. Hypertension also substantially accelerates coronary atherosclerosis and increases the risk of myocardial infarction by approximately threefold compared with normotensive individuals, partly through endothelial injury, oxidative stress, and dyslipidemia synergy. Cardiac arrhythmias, particularly atrial fibrillation, are 1.7 times more likely in hypertensive patients, related to left atrial enlargement from elevated filling pressures.

Cerebrovascular Complications

Hypertension is the single most important modifiable risk factor for stroke, responsible for approximately 54% of stroke cases globally. It increases the risk of both ischemic stroke (through accelerated large-artery atherosclerosis, cardioembolic sources from atrial fibrillation and LVH, and small-vessel lipohyalinosis of penetrating arteries causing lacunar infarcts) and hemorrhagic stroke (through rupture of Charcot-Bouchard microaneurysms in small perforating arteries subjected to chronically high pressure). Transient ischemic attacks (TIAs) share the same risk factor profile. Chronic hypertension also causes widespread cerebral small-vessel disease manifesting as white matter hyperintensities on MRI, lacunar infarcts, and cerebral microbleeds, cumulatively leading to vascular cognitive impairment and vascular dementia. The risk of dementia in hypertensive individuals is elevated approximately 60%, and treatment of hypertension in midlife is one of the most established strategies for reducing dementia risk in later life.

Renal Complications

The kidney is simultaneously a cause and a victim of hypertension. Hypertensive nephrosclerosis, resulting from arteriolar and glomerular damage, is a leading cause of end-stage renal disease (ESRD) in the United States, accounting for approximately 28% of incident ESRD cases (second only to diabetic nephropathy). Chronically elevated intraglomerular pressure (due both to systemic hypertension and to decreased arteriolar resistance from loss of autoregulatory capacity) leads to glomerular hyperfiltration, progressive glomerulosclerosis, tubular atrophy, and interstitial fibrosis. The earliest detectable renal manifestation is microalbuminuria (now termed "moderately increased albuminuria" — albumin-to-creatinine ratio 30–300 mg/g), which reflects increased glomerular permeability and is both a marker of systemic endothelial dysfunction and a predictor of progressive renal disease and cardiovascular events. Frank proteinuria (>300 mg/g or >0.5 g/day) indicates more advanced glomerular damage. CKD defined as GFR below 60 mL/min/1.73 m^2 or persistent albuminuria is present in approximately 10% of hypertensive patients in population studies.

Hypertensive Retinopathy

The retinal vasculature provides a unique window to observe the effects of chronic hypertension on the microcirculation. Hypertensive retinopathy is graded using the Keith-Wagener-Barker classification, which has four grades. Grade 1 shows mild arteriolar narrowing and increased arteriolar light reflex ("silver-wiring"). Grade 2 adds arteriovenous nicking (indentation of veins at arterial crossings, also called AV nipping) reflecting adventitial thickening. Grade 3 adds flame-shaped hemorrhages, cotton-wool spots (focal nerve fiber layer infarcts), and hard exudates (lipoprotein deposits from leaky capillaries). Grade 4 adds papilledema (optic disc swelling), which defines a hypertensive emergency involving the optic nerves. Grades 3 and 4 (malignant hypertension) are associated with fibrinoid necrosis of arterioles and indicate severely elevated pressure typically above 180/120 mmHg requiring emergency treatment.

Vascular Complications

Hypertension accelerates atherosclerosis throughout the arterial tree, increasing risk for peripheral artery disease (PAD) — reduced blood flow to the limbs, causing claudication, rest pain, and critical limb ischemia. The ankle-brachial index (ABI), the ratio of ankle to brachial systolic pressure, is a sensitive screen for PAD; a value below 0.9 indicates significant arterial stenosis. Hypertension is the strongest risk factor for aortic aneurysm formation and dissection. Chronic elevated pressure combined with medial degeneration predisposes to aortic aneurysm, most commonly in the infrarenal aorta (abdominal aortic aneurysm, AAA) but also in the thoracic aorta. Aortic dissection — a tear in the intima allowing blood to track through the media — occurs predominantly in the setting of severe, uncontrolled hypertension and Marfan syndrome, most commonly in the proximal aorta (Type A) or descending aorta (Type B). Hypertension is present in approximately 70–80% of aortic dissection cases.

9. Clinical Evaluation and Workup

A thorough clinical evaluation of a newly diagnosed hypertensive patient serves three purposes: confirming the diagnosis, identifying secondary causes, and stratifying total cardiovascular risk. The history should address the duration of hypertension, prior blood pressure readings, current and past antihypertensive medications, and adherence. Family history of hypertension, premature cardiovascular disease, renal disease, and endocrine tumors should be ascertained. Lifestyle factors including dietary sodium and potassium intake, alcohol consumption, physical activity, tobacco use, and recreational drug use (cocaine, amphetamines) should be reviewed. A thorough medication history is essential, specifically querying NSAIDs, oral contraceptives, decongestants, stimulants, steroids, and herbal supplements. Symptoms suggesting secondary causes should be sought: spells of headache, palpitations, diaphoresis (pheochromocytoma), weight gain, striae, muscle weakness (Cushing's), fatigue, cold intolerance, hair changes (hypothyroidism), daytime sleepiness, snoring, witnessed apneas (OSA), and flank pain or hematuria (renal causes).

Physical examination should include multiple blood pressure readings in both arms and, in younger patients or when coarctation is suspected, the lower extremities. Assessment of body mass index (BMI), waist circumference (central obesity threshold: >102 cm in men, >88 cm in women), and general habitus is important. Fundoscopic examination for hypertensive retinopathy grades should be performed. The cardiovascular examination should assess for LVH (laterally displaced apical impulse, S4 gallop), signs of heart failure (elevated JVP, rales, peripheral edema), and abnormal cardiac rhythms. Auscultation for bruits over the carotid arteries, renal arteries (epigastric and flank bruits), and femoral arteries addresses vascular disease and renovascular hypertension. The abdominal examination should note renal enlargement (polycystic kidneys) and aortic pulsatility. Neurological examination screens for prior stroke.

Baseline laboratory evaluation for all newly diagnosed hypertensives should include: urinalysis with microscopy (proteinuria, hematuria, casts suggesting renal parenchymal disease), urine albumin-to-creatinine ratio, complete metabolic panel (serum creatinine/estimated GFR, electrolytes, glucose, calcium), fasting lipid panel, thyroid-stimulating hormone (TSH), complete blood count, and an electrocardiogram (ECG) to assess for LVH and arrhythmias. Further testing for secondary hypertension is guided by clinical suspicion and should include plasma aldosterone-to-renin ratio if primary aldosteronism is suspected; plasma or 24-hour urine fractionated metanephrines for pheochromocytoma; morning serum cortisol and 24-hour urine cortisol or overnight dexamethasone suppression test for Cushing's; renal ultrasound or duplex Doppler for renovascular disease; polysomnography for OSA; and echocardiography for LVH, diastolic dysfunction, or when cardiac disease is a concern.

10. Lifestyle Management

Lifestyle modifications are the foundation of hypertension management and should be recommended to all patients regardless of whether pharmacologic treatment is initiated. They have the potential to lower systolic blood pressure by 4–11 mmHg when implemented rigorously, with additive effects when multiple interventions are combined.

Dietary Approaches

The Dietary Approaches to Stop Hypertension (DASH) diet is the most robustly evidence-supported dietary pattern for blood pressure reduction. The original DASH trial, published in the New England Journal of Medicine in 1997, compared three dietary patterns in 459 adults with elevated blood pressure and demonstrated that the DASH diet (rich in fruits, vegetables, low-fat dairy products, whole grains, poultry, fish, and nuts; low in red meat, sweets, and sugar-sweetened beverages; containing approximately 3,400 mg/day of potassium, 1,240 mg/day of calcium, and 500 mg/day of magnesium) reduced systolic blood pressure by 11.4 mmHg and diastolic by 5.5 mmHg in hypertensive subjects compared with the control diet. The DASH-Sodium trial subsequently demonstrated a dose-dependent relationship between sodium intake and blood pressure reduction, with the greatest reductions occurring when the DASH diet was combined with a low-sodium intake of approximately 1,500 mg/day, achieving a 8.9 mmHg reduction in systolic BP compared with a high-sodium (3,450 mg/day) control diet. Current guidelines generally recommend limiting sodium intake to less than 2,300 mg/day (approximately 1 teaspoon of table salt), though more aggressive targets of 1,500 mg/day are recommended for patients with hypertension, CKD, diabetes, or those who are Black.

Dietary potassium is blood pressure-lowering and cardioprotective. Increasing potassium intake to approximately 3,500–5,000 mg/day (achievable through generous consumption of fruits, vegetables, and legumes) is associated with a 4–5 mmHg reduction in systolic blood pressure. Potassium promotes natriuresis and modulates vascular tone. However, potassium supplementation must be used cautiously in patients with CKD or those taking RAAS inhibitors, where hyperkalemia is a risk. Magnesium and calcium may contribute modestly to blood pressure reduction through their roles in vascular smooth muscle function and are incorporated in the DASH diet. Moderate consumption of dark chocolate (high in flavanols) and dietary nitrates (found in green leafy vegetables and beets) have demonstrated modest blood pressure-lowering effects in clinical trials but are not primary recommendations.

Weight Loss and Physical Activity

For overweight or obese hypertensive patients, weight loss is one of the most effective lifestyle interventions. A reduction of 1 kg of body weight is associated with approximately 1 mmHg reduction in systolic blood pressure. The PREMIER trial demonstrated that comprehensive lifestyle modification achieving a mean weight loss of 4.9 kg lowered systolic blood pressure by 14 mmHg at 6 months. Aerobic physical activity — defined as at least 150 minutes per week of moderate-intensity activity (brisk walking, cycling, swimming) or 75 minutes of vigorous-intensity activity — lowers systolic blood pressure by approximately 5–8 mmHg and diastolic by 3–4 mmHg through mechanisms including reduced sympathetic nervous system activity, improved endothelial function, and favorable body composition changes. Dynamic resistance training (circuit training) also modestly reduces blood pressure, though isometric resistance exercise (as in some forms of strength training) can acutely elevate blood pressure substantially and requires caution in uncontrolled severe hypertension.

Alcohol and Tobacco

Alcohol reduction is recommended for hypertensive patients who drink excessively. Reducing alcohol intake from approximately 4 drinks per day to 1–2 drinks per day can lower systolic blood pressure by approximately 5.5 mmHg. Guidelines recommend limiting alcohol to no more than 2 standard drinks per day for men and 1 per day for women. Smoking cessation is mandatory in all hypertensive patients because tobacco is a major independent cardiovascular risk factor that compounds hypertension's effects dramatically. Each cigarette smoked transiently elevates BP by 5–10 mmHg for 30 minutes through nicotine-mediated catecholamine release. Chronic smoking accelerates atherosclerosis, endothelial dysfunction, and vascular stiffness.

11. Pharmacologic Treatment

When lifestyle modifications are insufficient to achieve blood pressure targets, or when cardiovascular risk is high, pharmacologic therapy is indicated. The 2017 ACC/AHA guidelines recommend pharmacologic treatment for all Stage 2 hypertension (≥140/90) and for Stage 1 hypertension (130–139/80–89) in patients with established cardiovascular disease or 10-year ASCVD risk ≥10%. Four major drug classes are considered first-line therapy: thiazide-type diuretics, angiotensin-converting enzyme (ACE) inhibitors, angiotensin receptor blockers (ARBs), and calcium channel blockers (CCBs). Beta-blockers, formerly a first-line option, are now primarily used in specific indications. Additional drug classes are available for patients requiring additional blood pressure lowering or with specific comorbidities.

Thiazide and Thiazide-like Diuretics

Thiazide diuretics inhibit the sodium-chloride cotransporter (NCC) in the distal convoluted tubule of the nephron, reducing sodium and water reabsorption. Hydrochlorothiazide (HCTZ, typically 12.5–25 mg/day) is the most commonly used, though chlorthalidone (12.5–25 mg/day) has greater evidence from trials (notably ALLHAT) and longer duration of action due to its prolonged tissue half-life (45–60 hours vs 6–15 hours for HCTZ). Indapamide is a thiazide-like agent with additional vasodilatory properties, commonly used in Europe. Key side effects include hypokalemia and hypomagnesemia (due to increased urinary potassium and magnesium losses), hyperuricemia and gout (through competitive inhibition of uric acid secretion), glucose intolerance (hypokalemia-mediated reduction in insulin secretion), mild hypercholesterolemia, hyponatremia (especially in elderly women), and sexual dysfunction. Thiazides are particularly effective in volume-sensitive or salt-sensitive hypertension and in Black patients. They are first-line for isolated systolic hypertension in the elderly. They are contraindicated in severe hypokalemia and gout.

ACE Inhibitors

ACE inhibitors block the conversion of angiotensin I to angiotensin II, reducing vasoconstriction and aldosterone secretion, leading to vasodilation and mild natriuresis. They also prevent the degradation of bradykinin, a vasodilatory peptide, which contributes to their efficacy but also causes their most common side effect — a dry, irritating, non-productive cough in approximately 10–20% of patients (higher rates in Asian populations, up to 40%). Commonly used agents include lisinopril (10–40 mg/day), enalapril (5–40 mg/day), ramipril (2.5–10 mg/day), and benazepril (10–40 mg/day). A potentially life-threatening side effect is angioedema (swelling of the lips, tongue, face, or larynx due to bradykinin accumulation), occurring in approximately 0.1–0.3% of patients — higher rates in Black patients. Other adverse effects include hyperkalemia and acute kidney injury, especially when combined with NSAIDs or in bilateral renal artery stenosis. ACE inhibitors are particularly beneficial in patients with diabetic nephropathy, CKD with proteinuria, heart failure with reduced ejection fraction, post-myocardial infarction, and high cardiovascular risk.

Angiotensin Receptor Blockers (ARBs)

ARBs block AT1 receptors directly, preventing angiotensin II effects including vasoconstriction and aldosterone secretion. Because they do not affect bradykinin metabolism, they do not cause cough and have a significantly lower risk of angioedema compared with ACE inhibitors (though angioedema has been reported rarely). ARBs share the renoprotective and cardiovascular benefits of ACE inhibitors and are the preferred alternative in patients who are intolerant of ACE inhibitors due to cough or angioedema. Commonly used agents include losartan (25–100 mg/day), valsartan (80–320 mg/day), olmesartan (10–40 mg/day), telmisartan (20–80 mg/day), and irbesartan (150–300 mg/day). Telmisartan has the longest half-life (~24 hours) among ARBs, providing 24-hour coverage with once-daily dosing. Key adverse effects include hyperkalemia and renal function decline, similar to ACE inhibitors. ACE inhibitors and ARBs should NOT be used together (dual RAAS blockade) due to increased risk of acute kidney injury, hyperkalemia, and hypotension without additional cardiovascular benefit, as demonstrated in the ONTARGET trial. Both classes are contraindicated in pregnancy (teratogenic — associated with oligohydramnios, fetal renal dysgenesis, skull hypoplasia, limb contractures).

Calcium Channel Blockers (CCBs)

CCBs block voltage-gated L-type calcium channels in vascular smooth muscle and cardiac cells, reducing intracellular calcium influx and thereby reducing smooth muscle contraction on agents in resistant hypertension), and heart failure with reduced ejection fraction. Key side effects of spironolactone include gynecomastia and sexual dysfunction (due to anti-androgenic effects), which are minimized with eplerenone (more selective MRA). Hyperkalemia is a risk, requiring monitoring, particularly in CKD. Alpha-1 adrenergic blockers (doxazosin, prazosin, terazosin) block vascular smooth muscle alpha-1 receptors, causing vasodilation. They are useful in patients with benign prostatic hyperplasia (BPH) for combined symptom management. Key adverse effect is first-dose orthostatic hypotension; they are generally reserved for add-on therapy or BPH comorbidity. Central alpha-2 agonists (clonidine, methyldopa, guanfacine) reduce sympathetic outflow from the central nervous system by stimulating inhibitory alpha-2 receptors in the rostral ventrolateral medulla. Methyldopa is the drug of choice for hypertension in pregnancy (see section 12). Clonidine is effective but limited by side effects including sedation, dry mouth, and rebound hypertension on abrupt discontinuation. Vasodilators — hydralazine and minoxidil — directly relax arteriolar smooth muscle. Hydralazine (particularly when combined with isosorbide dinitrate) reduces mortality in Black patients with heart failure and is used in pregnancy. Minoxidil is highly potent but causes significant reflex tachycardia and fluid retention, requiring concurrent use of a loop diuretic and a beta-blocker; it also causes hypertrichosis (excessive hair growth).

Combination Therapy

Most patients with Stage 2 hypertension will require two or more medications to achieve blood pressure control. The 2017 ACC/AHA guidelines recommend initiating two first-line agents from different drug classes if blood pressure is more than 20/10 mmHg above the target. Preferred combinations include a RAAS inhibitor (ACE inhibitor or ARB) plus a CCB (particularly amlodipine-based combinations, as demonstrated in ACCOMPLISH where amlodipine/benazepril was superior to benazepril/HCTZ in reducing cardiovascular events) or a RAAS inhibitor plus a thiazide diuretic, or a CCB plus a thiazide diuretic. The ACCOMPLISH trial also established that CCB/RAAS combinations are preferred over diuretic/RAAS combinations when possible. Fixed-dose combination (FDC) pills, combining two or three drugs in a single tablet, substantially improve adherence and are now strongly recommended when polypharmacy is required.

12. Special Populations

Hypertension in Diabetes

Hypertension and type 2 diabetes coexist in approximately 75% of diabetic patients and the combination dramatically multiplies cardiovascular risk. The blood pressure target for patients with diabetes is below 130/80 mmHg per most current guidelines. First-line agents preferred are ACE inhibitors or ARBs because of their demonstrated ability to reduce proteinuria and slow the progression of diabetic nephropathy beyond their blood pressure-lowering effect — a nephroprotective action mediated by preferential dilation of the efferent arteriole, reducing intraglomerular pressure. The UKPDS and HOT trials demonstrated that aggressive blood pressure lowering in diabetes significantly reduces microvascular (retinopathy, nephropathy) and macrovascular (stroke, MI) complications. ACE inhibitors and ARBs should not be combined (dual RAAS blockade increases adverse outcomes in diabetic patients, as shown in ONTARGET and VA NEPHRON-D).

Hypertension in Chronic Kidney Disease

CKD and hypertension exist in a bidirectional, mutually worsening relationship. The blood pressure target in CKD is below 130/80 mmHg. ACE inhibitors and ARBs are the cornerstone of antihypertensive therapy in CKD because they reduce proteinuria, slow nephron loss, and delay progression to ESRD beyond blood pressure-dependent effects. However, they can cause an initial 10–35% increase in serum creatinine due to reduced intraglomerular pressure (which represents hemodynamic adaptation, not true nephrotoxicity) and can precipitate hyperkalemia, particularly in advanced CKD (GFR below 30 mL/min). An increase in creatinine up to 30% from baseline is generally acceptable; greater increases warrant dose reduction or discontinuation. Thiazide diuretics lose efficacy when GFR falls below approximately 30 mL/min, at which point loop diuretics (furosemide, torsemide, bumetanide) are required for volume management.

Hypertension in Pregnancy

Hypertension in pregnancy encompasses several distinct conditions: chronic hypertension (predating pregnancy or diagnosed before 20 weeks), gestational hypertension (new-onset hypertension after 20 weeks without proteinuria or end-organ damage), preeclampsia (hypertension after 20 weeks with proteinuria $\geq$300 mg/24h or $>$300 mg/g albumin-to-creatinine ratio, or with end-organ damage in the absence of proteinuria), and the severe variant eclampsia (preeclampsia with seizures). HELLP syndrome (Hemolysis, Elevated Liver enzymes, Low Platelets) is a severe complication. Preeclampsia affects approximately 3–8% of pregnancies and is a leading cause of maternal and perinatal morbidity and mortality. The only definitive treatment for preeclampsia is delivery. Antihypertensive treatment in pregnancy aims to prevent maternal stroke and end-organ damage in severe hypertension ($\geq$160/110 mmHg) using labetalol (IV or oral), nifedipine (oral), or hydralazine (IV). Methyldopa remains safe and commonly used for chronic hypertension management throughout pregnancy. ACE inhibitors and ARBs are absolutely contraindicated in pregnancy due to teratogenicity. Magnesium sulfate is used for seizure prophylaxis and treatment in preeclampsia/eclampsia; low-dose aspirin (81–150 mg/day, initiated at 12–28 weeks) is recommended for preeclampsia prevention in high-risk patients. Blood pressure target in pregnancy is typically 120–160/80–105 mmHg, avoiding excessive lowering that could compromise uteroplacental perfusion.

Hypertension in Older Adults

Over 80% of adults above age 75 have hypertension, predominantly isolated systolic hypertension. Landmark trials including SHEP (Systolic Hypertension in the Elderly Program), Syst-Eur, and SPRINT (Systolic Blood Pressure Intervention Trial — which included adults over 75 in a prespecified subgroup) established that treatment of hypertension in older adults significantly reduces stroke, cardiovascular events, and all-cause mortality. SPRINT demonstrated that targeting SBP below 120 mmHg (intensive target) reduced cardiovascular events by 25% and or insufficient therapy (inadequate doses, inappropriate drug combinations, failure to include a diuretic), high dietary sodium intake overwhelming diuretic effects, volume overload from renal insufficiency or primary aldosteronism, OSA, and undetected secondary causes. The PATHWAY-2 trial established that spironolactone 25–50 mg/day is the most effective add-on treatment for resistant hypertension not already receiving an MRA, achieving a mean additional reduction of 8.7 mmHg compared with placebo. For patients who cannot tolerate spironolactone, amiloride (a potassium-sparing diuretic acting on ENaC), or eplerenone may be used. Other options for resistant hypertension include beta-blockers, alpha-blockers, clonidine, minoxidil, and hydralazine. Device-based therapies including renal denervation (catheter-based ultrasound or radiofrequency ablation of renal sympathetic nerves) and baroreflex activation therapy (stimulation of carotid sinus baroreceptors) are investigational but have shown modest benefits in randomized trials and are available in select centers for carefully chosen patients.

14. Hypertensive Urgency and Emergency

Hypertensive crisis refers to situations of severely elevated blood pressure (systolic greater than 180 mmHg and/or diastolic greater than 120 mmHg). It is subdivided into two clinically distinct entities based on the presence or absence of acute hypertension-mediated target-organ damage (HTOD).

Hypertensive urgency is defined as severely elevated blood pressure without evidence of acute target-organ damage. Symptoms may include headache, anxiety, or epistaxis. Despite the markedly elevated readings, there is no active end-organ injury, and rapid intravenous treatment is not necessary and may be harmful (precipitous drops in blood pressure can cause cerebral hypoperfusion and ischemic stroke). Management involves gradual oral blood pressure lowering over 24–48 hours using oral agents; a reasonable approach is to provide a dose of the patient's existing regimen, add a new oral agent (such as clonidine, amlodipine, or labetalol orally), and arrange close outpatient follow-up within 1–7 days. Patients should be observed and reassessed before discharge.

Hypertensive emergency is defined as severely elevated blood pressure with evidence of acute, progressive target-organ damage. This constitutes a true medical emergency requiring prompt hospital admission, intravenous antihypertensive therapy, and continuous blood pressure monitoring (typically with an intra-arterial line). Manifestations of hypertensive emergency include hypertensive encephalopathy (headache, confusion, visual disturbances, seizures, altered consciousness), acute ischemic or hemorrhagic stroke, aortic dissection, acute pulmonary edema with acute left ventricular failure, acute coronary syndrome (unstable angina/NSTEMI/STEMI), acute hypertensive nephrosclerosis (hematuria, proteinuria, red cell casts, rapidly rising creatinine), microangiopathic hemolytic anemia/thrombotic microangiopathy, and eclampsia/ severe preeclampsia.

The general principle for managing hypertensive emergency is controlled, stepwise reduction in blood pressure — typically targeting a 10–25% reduction in MAP within the first hour, then further gradual reduction to 160/100–110 mmHg within 2–6 hours, with further normalization over 24–48 hours. The critical exception is aortic dissection, where blood pressure should be lowered more rapidly (systolic to below 120 mmHg within the first hour) and heart rate controlled to below 60 beats per minute. The choice of IV agent depends on the specific manifestation: labetalol (IV bolus or infusion) and nicardipine (IV infusion) are commonly used versatile agents for most presentations; nitroprusside (IV infusion) is highly effective but requires careful monitoring due to cyanide and thiocyanate toxicity with prolonged use; nitroglycerin is preferred in pulmonary edema and acute coronary syndromes; esmolol and labetalol are preferred in aortic dissection; hydralazine IV in eclampsia; phentolamine in pheochromocytoma crisis. Oral clonidine may be used in urgency. The use of rapidly-acting sublingual nifedipine has been abandoned due to reports of uncontrolled hypotension and stroke.

15. Monitoring, Adherence, and Prognosis

After initiating antihypertensive therapy, patients should be followed up at 1 month to assess the blood pressure response, adherence, and side effects. Once blood pressure is at target, follow-up every 3–6 months is appropriate for stable patients. At each visit, blood pressure should be measured using correct technique; adherence should be assessed (pill counts, pharmacy refill records, or urine drug testing in suspected non-adherence); relevant laboratory tests should be repeated as needed (electrolytes, renal function, glucose, and urine albumin at least annually); and lifestyle factors should be reviewed and reinforced. The emergence of new symptoms suggesting target-organ damage or secondary causes warrants investigation.

Medication adherence is one of the most critical determinants of hypertension control and cardiovascular outcomes. Poor adherence — taking less than 80% of prescribed doses — is estimated to affect approximately 50% of hypertensive patients at 1 year and worsens over time. Factors associated with poor adherence include polypharmacy (multiple medications with complex regimens), side effects, cost, lack of symptoms (patients who feel well may not perceive the need for ongoing medication), poor patient-provider relationship, health literacy limitations, and socioeconomic barriers. Strategies to improve adherence include simplifying regimens to once-daily dosing, using fixed-dose combination pills, reducing out-of-pocket costs, patient education about the silent nature of hypertension and the importance of long-term treatment, home blood pressure monitoring (which improves engagement), and regular follow-up with reinforcement of lifestyle and medication goals.

The prognosis of hypertension is intimately tied to whether blood pressure is controlled, the presence of other cardiovascular risk factors, and the extent of existing target-organ damage at diagnosis. The relationship between blood pressure and cardiovascular mortality is log-linear and continuous; meta-analyses of prospective observational studies have shown that each 20 mmHg rise in SBP or 10 mmHg rise in DBP doubles the risk of stroke and ischemic heart disease mortality across the range of 115/75 to 185/115 mmHg. Conversely, treatment substantially reduces risk: meta-analyses of randomized controlled trials demonstrate that a 10 mmHg reduction in SBP reduces the risk of major cardiovascular events by approximately 20%, stroke by 27%, and heart failure by 28%. Over a lifetime, achieving blood pressure control can prevent more than 50% of all cardiovascular events attributable to hypertension.

The concept of total cardiovascular risk is central to modern hypertension management. Blood pressure does not act in isolation; it interacts multiplicatively with other risk factors including hyperlipidemia, diabetes, smoking, age, sex, family history, and existing vascular disease. Risk calculators such as the ACC/AHA Pooled Cohort Equations (estimating 10-year ASCVD risk), the Framingham Risk Score, and the ESC SCORE2 model integrate these factors to guide treatment intensity. A patient with Stage 1 hypertension and a 10-year ASCVD risk above 10% derives substantial absolute benefit from pharmacologic therapy, while a younger patient with isolated Stage 1 hypertension and a low ASCVD risk may achieve adequate risk reduction through lifestyle modification alone — at least for a period of time before disease progression occurs.

Long-term monitoring of hypertensive patients should include regular assessment of new target-organ damage: annual urine albumin-to-creatinine ratio and eGFR, periodic ECG or echocardiography in patients with symptoms or suspected LVH, and ophthalmologic evaluation for retinopathy. Cardiovascular risk reduction extends beyond blood pressure control to include concurrent management of other risk factors: statin therapy for LDL reduction in patients with cardiovascular disease or elevated 10-year risk, antiplatelet therapy where indicated, glycemic control in diabetes, tobacco cessation, and aspirin (no longer routinely recommended for primary prevention due to bleeding risk but essential for secondary prevention in established cardiovascular disease).

The overall prognosis for well-controlled hypertension in the modern era is substantially improved compared with undertreated disease. Patients who achieve and sustain blood pressure targets, engage in lifestyle modification, and receive regular monitoring can expect near-normal life expectancy and substantially reduced rates of stroke, myocardial infarction, heart failure, renal failure, and dementia. The public health challenge remains in identifying undiagnosed individuals, engaging treated but uncontrolled patients, and addressing the systemic barriers — cost, access, health literacy, systemic inequities — that perpetuate the large gaps between the evidence base and clinical reality seen globally in hypertension care.